Literature Review

Customer Churn Analysis and Retention Strategy for the Olist E-Commerce Marketplace

Project title	Customer Churn Analysis and Retention Strategy (Olist Brazilian E-Commerce dataset)
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Track	Data Engineering
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GitHub repository	https://github.com/nhahub/NHA-5-086
Document date	31 October 2026
Document version	1.0 (submitted for lecturer review)

1. Introduction

1.1 Purpose of this review

This review looks at published research that is related to our project: finding the patterns that lead to customer churn on the Olist marketplace and turning them into a dashboard and a retention strategy. The goal is not to summarize everything written about churn. The goal is to support the decisions we made in the Project Planning & Management document, namely how we define churn, which customer information we break churn down by, how we treat the class imbalance, how we avoid mistakes in the data preparation, and how we present the results.

1.2 Scope and approach

The review covers six themes: (1) how churn is defined when customers have no contract, (2) earlier work that used the Olist dataset, (3) factors linked to repeat purchase in e-commerce, (4) class imbalance and evaluation metrics, (5) data leakage and the observation window, and (6) dashboard design and retention strategy. Sources are peer-reviewed journal papers and books from marketing, operations and data mining, plus the Olist dataset page. Search terms included customer churn, non-contractual setting, e-commerce repeat purchase, Olist, class imbalance, data leakage and dashboard design. This is a focused review for a student project, not a systematic review.

2. Churn When There Is No Contract

In subscription businesses such as telecom, churn is visible because the customer cancels. Olist is different. It is a marketplace where customers simply buy or do not buy, and nobody announces that they have left. The literature calls this a non-contractual setting. In such settings the business cannot observe whether a customer is still "alive", so churn must be inferred from behavior (Schmittlein, Morrison & Colombo, 1987; Fader, Hardie & Lee, 2005).

Two ways to handle this appear in the literature:

- **Probabilistic models.** The Pareto/NBD model and its simpler version, the BG/NBD model, estimate the chance that a customer is still active from how recently and how often they bought (Schmittlein et al., 1987; Fader et al., 2005). These models need a purchase history with several transactions per customer.
- **A fixed inactivity window.** The analyst chooses a period, and a customer with no purchase in that period is labeled as churned. Studies on retail customers use this kind of definition, often together with a classification

model (Buckinx & Van den Poel, 2005). The weak point is that the window is a choice made by the analyst, so it has to be justified for the business.

How this affects our project. The Olist dataset has almost no repeat purchases (see Section 3), so there is too little purchase history per customer for the probabilistic models to work with. A fixed window is the simpler and more suitable option. Our 90-day rule from Section 1.4 of the planning document follows this approach. Because the window length is a choice, the optional 60-day and 180-day check in our plan is the right way to show how much the result depends on it.

3. Earlier Work on the Olist Dataset

3.1 Peer-reviewed study

The closest academic work to our project is Matuszelański and Kopczewska (2022), who studied churn in e-commerce using the public Olist data. They focused on whether a customer places a second order. Their main findings that matter to us are:

- Repeat buying is very rare. About 96.6% of the orders in their data came from customers on their first order, and only about 3.1% were second orders.
- The information available about a customer is limited, and the predictive signal is weak. Their best model, a gradient boosting model (XGBoost), reached a test AUC of about 0.65, while logistic regression reached about 0.59. The XGBoost training AUC was close to 1.0, so the large gap between training and test results points to overfitting.
- The product category of the first purchase added the most useful information among the extra variables they tested, more than review topics or regional population data.
- The product mix changes between the first and second order. For example, the share of bed, bath and table products rose from 11.4% to 13.8%, while watches and gifts fell from 6.8% to 3.8%.
- They grouped customer information into behavioral data (what was bought and paid), demographic or regional data, and perception data (review scores and review text).

3.2 Public analyses

Public notebooks and repositories on Olist report the same pattern of about 97% one-time buyers (for example, the Olist deep dive repository listed in the references). These are not peer reviewed, so we use them only as a sanity check on the numbers we will compute ourselves.

3.3 What this means for our project

- We should expect a very high share of churned customers and should report it openly, as planned in Risk 1.
- Because predictive signals are weak even for strong models, a dashboard that compares churn rates between customer groups is a sound choice. A prediction model stays optional, as in our scope.
- The planned breakdowns by product category, payment type, review score, delivery status and state are consistent with the variable groups used in the literature.
- Our label is stricter than "places a second order". A customer must place the second order within 90 days of the first, so the share of churned customers in our data is likely to be somewhat higher than 97%. We will confirm the exact figure in the exploration step.

4. Factors Linked to Repeat Purchase in E-Commerce

The literature suggests several groups of factors. The table shows how each group maps to the breakdowns planned for our dashboard.

Factor group	What the literature says	Breakdown in our project
Delivery and order fulfillment	Rao, Griffis and Goldsby (2011) link order fulfillment problems, such as late delivery or wrong items, to lower future purchasing by the affected customers.	Delivery delay, late-delivery flag

Factor group	What the literature says	Breakdown in our project
Customer perception	Matuszelański and Kopczewska (2022) treat review ratings and review text as a separate group of variables that describe how the customer judged the experience.	Review score
First purchase	Miguéis, Van den Poel, Camanho and e Cunha (2012) show that the product categories bought first carry information about later partial churn. The Olist study above found the same for first-order category.	Product category, number of items, order value, freight cost
Payment behavior	Payment value and shipping cost are standard behavioral variables in the Olist churn study.	Payment type, installments, freight cost
Location	Regional information was tested by Matuszelański and Kopczewska (2022). It added little compared with first-order information, but location is still a common business segment.	Customer state

Ngai, Xiu and Chau (2009) reviewed data mining in customer relationship management and list customer retention as one of its main application areas. This supports the idea that a retention analysis based on transaction data is a standard and accepted type of project.

5. Class Imbalance and Evaluation Metrics

When one class is much larger than the other, a model can reach a very high accuracy by always predicting the large class, while being useless in practice (He & Garcia, 2009). The literature offers two kinds of solutions: change the data, for example by resampling or by creating synthetic examples such as with SMOTE (Chawla, Bowyer, Hall & Kegelmeyer, 2002), or change the learning method, for example by giving more weight to the small class (He & Garcia, 2009).

For evaluation, Saito and Rehmsmeier (2015) show that the precision-recall plot gives a more informative picture than the ROC curve when the classes are strongly imbalanced. In churn work, lift is also common: it compares how many real positives a model finds in its top-ranked group, such as the top 5% of customers, against random selection (Matuszelański & Kopczewska, 2022). Verbeke, Dejaeger, Martens, Hur and Baesens (2012) go one step further and argue that models should be judged by the profit of the retention action, not only by statistical scores.

How this affects our project. In most churn studies the churned customers are the small class. In Olist the opposite is true: churn = 1 is the large class, and the repeat buyers (churn = 0) are the rare ones. If the optional model is built, we will use class weights and report Precision, Recall, F1-score and ROC-AUC, as planned in Risk 1, and we will treat the repeat buyers as the class of interest when reading Precision and Recall. In the dashboard we will compare churn rates between groups instead of relying on one overall number.

6. Data Leakage and the Observation Window

Data leakage means that the analysis uses information that would not be known at the moment the prediction or decision is made. Kaufman, Rosset, Perlich and Stitelman (2012) explain that leakage is easy to introduce during data preparation and gives results that look better than they really are. In our project this could happen if a breakdown used information from after the first order, such as the second order itself. Our plan limits the breakdowns to information known from the first order (Risk 4).

One detail needs care. A review score is written after delivery, so the team should check in the data that each review was created before the 90-day period closes and that it belongs to the first order, not a later one. QA will include this in the data tests.

The second issue is the observation window. A customer whose first order happened less than 90 days before the end of the dataset has not had the full 90 days to come back, so labeling them as churned would be unfair. Survival analysis calls this right-censoring. Instead of modeling it, our plan removes these customers from the analysis (Risk 3), which is the simpler option. Survival analysis, which handles censored customers directly, remains a possible future extension, but it is outside our scope.

7. Dashboards and Retention Strategy

7.1 Dashboard design

Few (2006) describes a dashboard as a way to monitor the most important information at a glance, ideally on one screen. His main recommendations are to choose the few measures that matter, to use simple charts such as bars and lines, to avoid decoration and to keep the layout consistent. Our planned design follows this: KPI cards at the top, a small set of charts for the main drivers, and filters that update all charts together.

7.2 Retention strategy

Reichheld and Sasser (1990) argue that keeping customers is more profitable than it looks, because the cost of acquiring a customer is paid once while the profit from a retained customer continues. Gupta et al. (2006) review models of customer lifetime value and show that the retention rate is one of its main drivers. Together, these works justify the last business KPI in our plan, the expected benefit of the proposed retention actions. When only a few customers return, even a small increase in the repeat rate can matter, so the strategy report should show the size of each segment as well as its churn rate.

8. Summary and Gap

Theme	Main point from the literature	Decision in our project
Churn definition	Without a contract, churn is inferred from inactivity; the window is an analyst choice.	90-day rule, with optional 60 and 180-day check
Olist data	About 97% of customers buy once; predictive signal is weak.	Group comparison dashboard; model is optional
Drivers	Delivery problems, review score, first-order category and payment details are linked to repeat buying.	Breakdowns by delivery, review, category, payment, state
Imbalance	Accuracy misleads; use precision, recall, F1, ROC-AUC or precision-recall.	Imbalance report; class weights if a model is built
Leakage and censoring	Use only information available at the decision point; handle customers observed for too short a time.	First-order information only; remove customers with less than 90 days
Dashboard and strategy	Simple, focused dashboards; retention has a clear financial value.	KPI cards, charts, filters; expected benefit of actions

Gap. The Olist churn study focuses on predicting a second order with machine learning models. Our project complements it by building a customer-level churn label with a documented rule, explaining the drivers with simple and testable breakdowns, and linking them to retention actions in a dashboard that a non-technical business user can read.

Limitations of this review

- Only one peer-reviewed study based on the Olist dataset was used; most other public work is not peer reviewed.
- The dataset covers 2016 to 2018 and only the Brazilian market, so findings may not hold for other countries or later years (see Risk 13).
- The review is focused, not exhaustive.

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